



School of Computer & Systems Sciences

MCA Ist Semester

Programming in C, CS 401-End Semester Exam Monsoon

22-4-2022, 11 AM to 2 PM

NOTE: Questions 1 to 5 carry 2 marks each, **All are mandatory**

(2*5 =10 marks)

Q1. C language was evolved from -

- a) A
- b) B
- c) D
- d) E

Q2. C was mainly developed for –

- a) System Programming
- b) Logic Programming
- c) Network Programming
- d) Application Programming

Q3. The most complex part of C is to work with...?

- a) Loops
- b) control structures
- c) pointers
- d) structures

Q4. Beside C the best networking language is?

- a) Prolog
- b) Java
- c) pascal
- d) Perl

Q5. C is an AI language?

- a) True
- b) False

Note Question 6 to 11 carry 5 marks each, Attend **any 4 questions** (4 * 5 = 20 marks)

Q6. Discuss the Fundamental differences between generation of Programming Languages?

Q7. Describe the working of bitwise operators in C?

Q8. Discuss the uses of Assignment operators in C?

Q 9. According to your understanding which condition statement is better to use and why: nested IF or Switch case?

Q 10. What is the advantage of do-while loop, support your answer with any example?

Q 11 According to your observation which is better to use, Structures or Unions and why?

Note Question 12 to 14 carry 10 marks each, Attend **any 2 questions** (2 * 10 = 20 marks)

Q 12. Describe Levels of Programming Languages in details with their advantages and disadvantages, using appropriate examples?

Q 13. Write down the complete history of C language as a state of art in your own language?

Q 14. Why C is considered as the most powerful programming language till date, discuss in detail, describe the tools and functions which makes C unique from other programming languages?



School of Computer & Systems Sciences

MCA Ist Semester

Programming in C, CS 401-End Semester Exam Monsoon

1-3-2023, 10 AM to 1 PM

NOTE: Questions 1 to 5 carry 2 marks each, All are mandatory marks)

(2*5 =10

Q1. C language was evolved in -

- a) 1940
- b) 1950
- c) 1960
- d) 1970

Q2. C was mainly developed for –

- a) System Programming
- b) Logic Programming
- c) Network Programming
- d) Application Programming

Q3. C is a ...?

- a) Low level programming language
- b) Middle level programming language
- c) High level programming Language
- d) AI programming language

Q4. Total number of 8 Bit ASCII characters are?

- a) 128
- b) 255
- c) 256
- d) 1024

Q5. Which of the following is not a key word in C?

- a) Char
- b) Int
- c) Real
- d) Float

Note Question 6 to 11 carry 5 marks each, Attend any 4 questions (4 * 5 = 20 marks)

Q6. Discuss the working of files made from editing to execution of a C program?

Q7. Discuss the use of Linkers and Loaders of C compiler?

Q8. Write down the structure of a C program with appropriate instructions?

Q9. Write the difference and working of the following functions strcat(), strcmp(), strcpy(), strlen() ?

Q10. What is the advantage of for loop, support your answer with any example?

Q11 According to your observation why Pointers are such a powerful tool of C and why?

Note Question 12 to 14 carry 10 marks each, Attend **any 2 questions** (2 * 10 = 20 marks)
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Q12. What are header files in C, Why and how they are used, discuss the difference and similarities between stdio.h, stdlib.h, conio.h?

Q13. Write down the need and uses of Structure and Union in C, also discuss the similarities and Differences between the both?

Q14. Why C is considered as the most powerful programming language till date, discuss in detail, describe the tools and functions which makes C unique from other programming languages?

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Programming in C (CS-401)
End Semester Exam
Date: 18.12.2024@10am

Total Marks: 50

Time: 3hr

All Compulsory

1. Wrote note on
 - a. Software types
 - b. Machine and assembly language
 - c. Call by value and call by reference in function
 - d. Dynamic memory allocation function3*4=12
2. Write a program to print if negative number is entered. 5
3. Multiply two positive numbers without * operator. 5
4. Program to add the elements of an array, where array size is 10. 5
5. Write a program to find the length of a string. 5
6. Program to understand postfix/prefix increment /decrement in a pointer variable of the base integer. 5
7. Program to write Roll number and marks obtained by a student side-by-side in a matrix using 2D array. 5
8. What are local and global variables? Explain with defining a problem and writintg a C program. 8

School of Computer and Systems Sciences,
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New Delhi -110067
Programming in C (CS-401)
End semester Exam
Date: 12.12.2025

Total Marks:50

Time: 3hr

1. Write note on:

- a. Struct and union
- b. Preprocessor directives
- c. Keywords in C
- d. Malloc and calloc
- e. Address operator with examples

4x5=20

2. Program to print the sum of digits of a user input number using function. 6
3. Write a program to check if the number elements is even or odd using functions 6
4. Write a program to check whether a person is eligible to cast a vote. Take input as year of birth from the user. Also, check for a negative age. 6
5. Write a program to calculate the electricity bill. Given units: 6
 - first 100 units → ₹10/unit
 - next 100 → ₹14/unit
 - above 200 → ₹20/unit

Calculate the total amount using a function.

6. Write a C program to insert a number entered by the user into an array at a user-defined position. The program should:
 - Take the size of the array and its elements as input from the user.
 - Ask the user for the number to be inserted.
 - Ask for the position where the number should be inserted.
 - Shift the elements accordingly and display the updated array.

OR

Write a program to create two user-defined arrays 1 and 2, take size and elements of array from user. Merge the array 1 and 2 into a third array and print the third array. 6